

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL2- ARTICLE REVIEW

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Code of Conduct:

I Ganji Madhan Kumar(192472328)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:G.Madhan

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Task 1: Article Summary

Full Citation:

- Authors: Z. Wang, X. Liu, and Y. Zhang
- Title: Adaptive and Efficient Resource Allocation in Cloud Datacenters Using Actor-Critic Deep Reinforcement Learning
- Journal: IEEE Transactions on Services Computing
- Year: 2022
- DOI / Link: <https://doi.org/10.1109/TSC.2021.3135652>
- Domain / Application Area: Cloud Computing Infrastructure & Resource Management
- Specific ML Problem: Multi-objective continuous resource allocation (CPU, memory, storage) and job scheduling under dynamic, high-dimensional state spaces using Deep Reinforcement Learning (DRL).

Key Objective Summary:

In multi-tenant cloud data centers, fluctuating user workloads and heterogeneous hardware infrastructure make optimal resource allocation challenging. Traditional rule-based, heuristic (e.g., Shortest Job First), and value-based reinforcement learning (e.g., Deep Q-Networks) techniques fail to handle high-dimensional, continuous action spaces. Value-based methods suffer from action space explosion, while simple policy gradient methods suffer from high variance and poor sample efficiency during training.

To bridge this research gap, the authors proposed an Asynchronous Advantage Actor-Critic (A3C) Deep Reinforcement Learning framework. The goal was to dynamically adjust resources to simultaneously minimize job latency, lower job dismissal rates (SLA violations), and maximize server energy efficiency. By combining an actor network (to directly output scheduling policies across continuous spaces) and a critic network (to estimate state-value functions and reduce policy gradient variance via an advantage function), the proposed framework continuously learns optimal allocation strategies without needing explicit prior statistical models of workload behavior.

Task 2: Methodology Analysis

ML Techniques & Dataset Description:

- ML Technique: The paper employs the Asynchronous Advantage Actor-Critic (A3C) algorithm, a model-free policy-based Deep Reinforcement Learning architecture:
 1. State Space (S): Includes current server CPU/memory/storage utilization, active queue capacity, and incoming job characteristics (duration, resource requirements).
 2. Action Space (A): Continuous mapping decisions assigning incoming jobs to specific target servers and allocating specific quotas of CPU, memory, and bandwidth.
 3. Reward Function (R): A composite function balancing latency minimization, SLA breach penalties, and total physical server energy consumption.
- Dataset:
 1. Source: Real-world cluster trace logs from Google Cloud Data Centers (Google Cluster Workload Traces).
 2. Size & Features: Millions of job request events recorded across 29 days. Features extracted include Job ID, Submit Time, CPU Requested, Memory Requested, Disk Space Requested, and Job Duration.

3. Data Preparation: Min-Max normalization to scale resource demands into standard ranges [0,1], time-series aggregation into discrete scheduling intervals, and removal of corrupted/missing trace logs.

Course Topic Mapping & Methodological Evaluation:

- Course Topic Mapping: Course Outcome 4 – Reinforcement Learning (RL) (specifically Policy Gradient methods, Actor-Critic Architectures, and Markov Decision Processes).
- Methodological Strength: The use of an Advantage Function $A(s, a) = Q(s, a) - V(s)$ inside the Actor-Critic design significantly stabilizes policy gradients and speeds up convergence compared to baseline Deep Q-Networks (DQN) or raw REINFORCE policy gradients in complex high-dimensional environments.
- Methodological Limitation: High dependence on simulator fidelity. Training the DRL agent requires millions of step interactions; training it offline using static historical trace logs can induce off-policy distribution shifts when deployed live in real-time cluster systems.

Task 3: Results & Critical Evaluation

Key Benchmark Results:

Algorithm / Baseline	Average Job Latency (s)	Job Dismissal Rate (SLA Violations)	Relative Energy Consumption (%)
Random Allocation	148.5	12.4%	100.0%
Shortest Job First (SJF)	92.1	8.2%	88.4%
Longest Job First (LJF)	115.3	9.6%	91.2%
Deep Q-Network (DQN)	64.2	4.1%	76.8%
Proposed A3C DRL Model	41.8	1.2%	64.5%

Critical Evaluation of Results:

- Realism & Biases: The reported metrics show significant performance improvements (e.g., latency dropping to 41.8 seconds). However, these evaluations rely heavily on trace-driven simulations where network latency and hardware degradation factors are simplified. Additionally, the dataset bias toward Google's specific batch workloads might obscure how the model responds to spike-heavy, real-time web traffic workloads.
- Strengthening Experiments: The evaluation would be stronger if evaluated against extreme stress-testing scenarios (e.g., sudden DDoS-like traffic spikes) and out-of-distribution workload patterns, alongside measuring total computational overhead incurred by inference latency of the deep neural network itself.

Real-World Applicability:

- **Deployment:** Applicable in enterprise public/private cloud infrastructure (AWS, Azure, OpenStack cluster managers) and Edge-Computing networks where resource constraints demand real-time adaptation.
- **Scaling Challenges:**
 1. *Data & Cold-Start:* Training DRL models requires extensive interaction data; poor early-stage actions ("exploration phase") in production can cause service outages.
 2. *Inference Latency:* Evaluating deep neural network passes within millisecond-level scheduling windows adds computational overhead to the master node.
 3. *Safety & Ethics:* Algorithmic starvation of low-priority tenant tasks in favor of resource-heavy tenants without strict guardrails.

Task 4: Reflection & Learning Outcome

Key Insights & Syllabus Connections:

1. **Exploration vs. Exploitation Trade-off in Continuous Spaces:** Learning how stochastic policies generated by policy-gradient Actor-Critic methods handle high-dimensional action spaces better than value-iteration methods (ϵ -greedy DQN) directly reinforces the Reinforcement Learning / MDP module.
2. **Reward Function Engineering:** Understanding that complex multi-objective goals (energy efficiency vs. response time) must be mathematically transformed into a scalar reward function mapped directly to the fundamental concept of Markov Decision Process (MDP) Optimization.
3. **Handling Non-Stationary Environments:** Observing how reinforcement learning agents handle dynamically changing workload arrival distributions deepened my understanding of Inductive Bias and Generalization across statistical ML models.